Homework 4 Solutions Discussion

STATS 220A

Q1: Setup and Statement

Let N be a nonneg integer-valued r.v. with $\mathbb{E}[N^2] < \infty$. Let $X_1, X_2, \ldots$ be i.i.d., nonneg, independent of N, with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Define $S_N = X_1 + \cdots + X_N$ (with $S_0 = 0$).

(a) Compute $\mathbb{E}[S_N \mid N]$ and use the tower property to show $\mathbb{E}[S_N] = \mu\,\mathbb{E}[N]$.

(b) Use the law of total variance to show $\mathrm{Var}(S_N) = \sigma^2\,\mathbb{E}[N] + \mu^2\,\mathrm{Var}(N)$.

(c) If $N \sim \mathrm{Poisson}(\lambda)$, show $\mathrm{Var}(S_N) = \lambda\,\mathbb{E}[X_1^2]$.

(d) Branching process: prove $\mathbb{E}[Z_n] = m^n$ and $\mathrm{Var}(Z_n) = v^2 m^{n-1} \dfrac{m^n - 1}{m - 1}$ for $m \neq 1$.

Q1(a): Computing $\mathbb{E}[S_N \mid N]$

The key is to condition on the event $\{N = n\}$.

Given $N = n$, the sum $S_N = X_1 + \cdots + X_n$ is a *deterministic* function of the i.i.d. sequence $X_1, \ldots, X_n$, each with mean μ. So

$$\mathbb{E}[S_N \mid N = n] = n\mu.$$

Since this holds for every n, the conditional expectation (as a random variable) is

$$\mathbb{E}[S_N \mid N] = N\mu.$$

Why is this immediate? Once N is revealed, the randomness in the X_i's averages out to μ per term, and N counts the number of terms.

Q1(a): Tower property gives Wald's identity

By the tower (iterated expectation) property,

$$\mathbb{E}[S_N] = \mathbb{E}\big[\mathbb{E}[S_N \mid N]\big] = \mathbb{E}[N\mu] = \mu\,\mathbb{E}[N].$$

Remark. This is Wald's identity. It requires only $\mathbb{E}[N] < \infty$ and $\mathbb{E}[X_1] = \mu < \infty$; the finite-second-moment assumption is needed only in part (b).

Q1(b): Law of total variance — setup

Recall the *law of total variance*:

$$\mathrm{Var}(S_N) = \mathbb{E}\big[\mathrm{Var}(S_N \mid N)\big] + \mathrm{Var}\big(\mathbb{E}[S_N \mid N]\big).$$

We compute each term separately.

Q1(b): First term — $\mathbb{E}[\mathrm{Var}(S_N \mid N)]$

Given $N = n$, the X_i are i.i.d. with variance σ^2, so

$$\mathrm{Var}(S_N \mid N = n) = n\sigma^2, \qquad \text{hence} \qquad \mathrm{Var}(S_N \mid N) = N\sigma^2.$$

Taking expectation:

$$\mathbb{E}\big[\mathrm{Var}(S_N \mid N)\big] = \sigma^2\,\mathbb{E}[N].$$

Interpretation. This term captures randomness in S_N due to the X_i's, *given* how many terms there are.

Q1(b): Second term — $\mathrm{Var}(\mathbb{E}[S_N \mid N])$

From part (a), $\mathbb{E}[S_N \mid N] = \mu N$. Therefore

$$\mathrm{Var}\big(\mathbb{E}[S_N \mid N]\big) = \mathrm{Var}(\mu N) = \mu^2 \,\mathrm{Var}(N).$$

Interpretation. This term captures randomness in S_N due to N itself: even if each X_i equaled its mean μ, the sum would still be random because N is random.

Combining both terms:

$$\mathrm{Var}(S_N) = \sigma^2 \,\mathbb{E}[N] + \mu^2 \,\mathrm{Var}(N). \qquad \square$$

Q1(c): Poisson N — $\mathrm{Var}(S_N) = \lambda\,\mathbb{E}[X_1^2]$

For $N \sim \mathrm{Poisson}(\lambda)$ we have $\mathbb{E}[N] = \mathrm{Var}(N) = \lambda$ (the Poisson is its own "variance equals mean" distribution).

Substitute into part (b):

$$\mathrm{Var}(S_N) = \sigma^2 \cdot \lambda + \mu^2 \cdot \lambda = \lambda(\sigma^2 + \mu^2).$$

But $\sigma^2 + \mu^2 = \mathrm{Var}(X_1) + (\mathbb{E}[X_1])^2 = \mathbb{E}[X_1^2]$ (the second moment identity). Hence

$$\mathrm{Var}(S_N) = \lambda\,\mathbb{E}[X_1^2]. \qquad \square$$

Remark. The collapse $\sigma^2 + \mu^2 = \mathbb{E}[X_1^2]$ is special to the Poisson: the two sources of variance merge into a single second moment because $\mathbb{E}[N] = \mathrm{Var}(N)$.

Q1(d): Branching process — structural observation

Let $Z_0 = 1$ and $Z_{n+1} = Y_1^{(n)} + \cdots + Y_{Z_n}^{(n)}$, where the offspring $\{Y_i^{(n)}\}$ are i.i.d. with mean $m > 0$ and variance $v^2 < \infty$, independent of Z_n.

Key observation. Z_{n+1} given $Z_n = k$ is precisely a random sum of the type in parts (a)–(b), with the number of summands equal to k (deterministic once we condition) and each summand $Y_i^{(n)}$ playing the role of X_i.

This structural match is what makes the induction work cleanly.

Q1(d)(i): Proving $\mathbb{E}[Z_n] = m^n$ by induction

Base case. $\mathbb{E}[Z_0] = 1 = m^0$. ✓

Inductive step. Assume $\mathbb{E}[Z_n] = m^n$.
Condition on Z_n: given $Z_n = k$, the offspring sum $Z_{n+1} = Y_1^{(n)} + \cdots + Y_k^{(n)}$ has mean km by linearity. So

$$\mathbb{E}[Z_{n+1} \mid Z_n = k] = km, \qquad \text{i.e.} \qquad \mathbb{E}[Z_{n+1} \mid Z_n] = mZ_n.$$

By the tower property,

$$\mathbb{E}[Z_{n+1}] = \mathbb{E}\big[\mathbb{E}[Z_{n+1} \mid Z_n]\big] = m\,\mathbb{E}[Z_n] = m \cdot m^n = m^{n+1}. \qquad \square$$

Q1(d)(ii): Variance formula — base case and setup

Claim. For $m \neq 1$,

$$\mathrm{Var}(Z_n) = v^2 m^{n-1} \frac{m^n - 1}{m - 1}.$$

Base case $n = 1$**.** Z_1 is a single family of offspring from $Z_0 = 1$, so $\mathrm{Var}(Z_1) = v^2$. The formula gives $v^2 \cdot m^0 \cdot (m-1)/(m-1) = v^2$. ✓

Inductive step. Assume the formula holds at generation n. Apply the law of total variance:

$$\mathrm{Var}(Z_{n+1}) = \mathbb{E}\big[\mathrm{Var}(Z_{n+1} \mid Z_n)\big] + \mathrm{Var}\big(\mathbb{E}[Z_{n+1} \mid Z_n]\big).$$

Q1(d)(ii): Inductive step — first term

Given $Z_n = k$, the offspring $Y_1^{(n)}, \ldots, Y_k^{(n)}$ are i.i.d. with variance v^2, so $\mathrm{Var}(Z_{n+1} \mid Z_n = k) = kv^2$. Hence

$$\mathrm{Var}(Z_{n+1} \mid Z_n) = v^2 Z_n.$$

Taking expectation and using $\mathbb{E}[Z_n] = m^n$:

$$\mathbb{E}\big[\mathrm{Var}(Z_{n+1} \mid Z_n)\big] = v^2\,\mathbb{E}[Z_n] = v^2 m^n.$$

Q1(d)(ii): Inductive step — second term and conclusion

From (d)(i), $\mathbb{E}[Z_{n+1} \mid Z_n] = mZ_n$. So

$$\mathrm{Var}\big(\mathbb{E}[Z_{n+1} \mid Z_n]\big) = \mathrm{Var}(mZ_n) = m^2\,\mathrm{Var}(Z_n) = m^2 \cdot v^2 m^{n-1}\,\frac{m^n - 1}{m - 1}.$$

Combining both terms:

$$\begin{aligned}
\mathrm{Var}(Z_{n+1}) &= v^2 m^n + v^2 m^{n+1}\,\frac{m^n - 1}{m - 1} \\
&= v^2 m^n \left(\frac{m - 1 + m^{n+1} - m}{m - 1} \right) \\
&= v^2 m^n \cdot \frac{m^{n+1} - 1}{m - 1}. \qquad \square
\end{aligned}$$

This matches the claimed formula with n replaced by $n + 1$.

Q2: Setup and Statement

Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_n = 1) = p_n$ and $\mathbb{P}(X_n = 0) = 1 - p_n$.

(a) $p_n = 1/n$: show $X_n \to 0$ in L^1 but $X_n \not\to 0$ a.s.

(b) $p_n = 1/n^2$: show $X_n \to 0$ a.s.

(c) $p_n = c/n$ for fixed $c > 0$ and $S_n = X_1 + \cdots + X_n$:

  (i) Show $\mathbb{E}[S_n] \sim c \log n$.

  (ii) Show $\mathrm{Var}(S_n) \leq c(\log n + 1)$.

  (iii) Prove $S_n / \log n \to c$ in probability.

Q2(a): L^1 convergence when $p_n = 1/n$

Since $X_n \in \{0, 1\}$, we have $|X_n| = X_n$, so

$$\mathbb{E}[|X_n|] = \mathbb{E}[X_n] = p_n = \frac{1}{n} \to 0.$$

By definition, $X_n \to 0$ in L^1.

Remark. L^1 convergence implies convergence in probability (Markov's inequality), but does *not* imply almost sure convergence. Part (a) makes this gap explicit.

Q2(a): Failure of a.s. convergence — Borel–Cantelli II

Let $A_n = \{X_n = 1\}$. The events A_n are *independent* because the X_n are independent.

Compute:

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

By the **second Borel–Cantelli lemma** (independence + divergent series $\Rightarrow$ i.o. a.s.):

$$\mathbb{P}(X_n = 1 \text{ i.o.}) = 1.$$

On this probability-1 event, X_n returns to 1 infinitely often and therefore cannot converge to 0. So $X_n \not\to 0$ a.s.

Q2(b): a.s. convergence when $p_n = 1/n^2$ — Borel–Cantelli I

Let $A_n = \{X_n = 1\}$.

Compute:

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty.$$

By the **first Borel–Cantelli lemma** (summable series $\Rightarrow$ only finitely many events occur a.s., no independence required):

$$\mathbb{P}(X_n = 1 \text{ i.o.}) = 0.$$

Therefore $X_n = 0$ for all sufficiently large n a.s., so $X_n \to 0$ a.s.

Contrast with (a). The dividing line is convergence vs. divergence of $\sum p_n$. For independent Bernoullis, the two BC lemmas give a complete dichotomy.

Q2(c): Setup — a remark on $c > 1$

Fix $c > 0$ and set $p_n = c/n$.

Remark on $c > 1$. For small n, $c/n > 1$, which is not a valid probability. Interpret $p_n = \min(1, c/n)$, or restrict to $n \geq \lceil c \rceil$ where $c/n \leq 1$. Since only finitely many initial terms are affected, all three asymptotic results below are unchanged.

Let $H_n := \sum_{k=1}^{n} 1/k$ denote the nth harmonic number. The key estimate is $H_n = \log n + \gamma + o(1)$, in particular $H_n \leq \log n + 1$.

Q2(c)(i): $\mathbb{E}[S_n] \sim c \log n$

By independence and linearity,

$$\mathbb{E}[S_n] = \sum_{k=1}^{n} \mathbb{E}[X_k] = \sum_{k=1}^{n} p_k = c \sum_{k=1}^{n} \frac{1}{k} = cH_n.$$

Dividing by $\log n$ and using $H_n / \log n \to 1$:

$$\frac{\mathbb{E}[S_n]}{\log n} = \frac{cH_n}{\log n} \to c, \qquad \text{i.e.} \qquad \mathbb{E}[S_n] \sim c \log n. \qquad \square$$

Q2(c)(ii): $\mathrm{Var}(S_n) \leq c(\log n + 1)$

The X_k are independent, so

$$\mathrm{Var}(S_n) = \sum_{k=1}^{n} \mathrm{Var}(X_k) = \sum_{k=1}^{n} p_k(1 - p_k).$$

Since $p_k(1 - p_k) \leq p_k$ for any $p_k \in [0, 1]$,

$$\mathrm{Var}(S_n) \leq \sum_{k=1}^{n} p_k = cH_n \leq c(\log n + 1). \qquad \square$$

In particular, $\mathrm{Var}(S_n) = O(\log n)$.

Q2(c)(iii): $S_n/\log n \to c$ in probability

Set $T_n := S_n/\log n$. The strategy: show $T_n \to c$ in L^2, which implies convergence in probability.

Mean. From (i),

$$\mathbb{E}[T_n] = \frac{\mathbb{E}[S_n]}{\log n} = \frac{cH_n}{\log n} \to c.$$

Variance. From (ii),

$$\mathrm{Var}(T_n) = \frac{\mathrm{Var}(S_n)}{(\log n)^2} \leq \frac{c(\log n + 1)}{(\log n)^2} \to 0.$$

Q2(c)(iii): Completing the argument

Compute the L^2 error:

$$\mathbb{E}\big[(T_n - c)^2\big] = \mathrm{Var}(T_n) + \big(\mathbb{E}[T_n] - c\big)^2.$$

Both terms $\to 0$, so $T_n \to c$ in L^2, hence in probability:

$$\frac{S_n}{\log n} \xrightarrow{\mathbb{P}} c. \qquad \square$$

Remark. The L^2 route is clean here because the variance bound from (ii) directly controls the L^2 fluctuations. Alternatively, Chebyshev applied to T_n gives the same conclusion without explicitly invoking the L^2 hierarchy:

$$\mathbb{P}\big(|T_n - c| > \varepsilon\big) \le \frac{\mathrm{Var}(T_n) + (\mathbb{E}[T_n] - c)^2}{\varepsilon^2} \to 0.$$

Final Checklist

- **Q1(a):** Condition on N; tower property yields Wald's identity $\mathbb{E}[S_N] = \mu\mathbb{E}[N]$.
- **Q1(b):** Law of total variance decomposes into "X-variability" + "N-variability".
- **Q1(c):** Poisson has $\mathbb{E}[N] = \mathrm{Var}(N) = \lambda$; second moment identity $\sigma^2 + \mu^2 = \mathbb{E}[X_1^2]$ collapses the formula.
- **Q1(d):** Branching = random sum; induction with LTV; algebra telescopes for the variance.
- **Q2(a):** $L^1 \to 0$ trivial; a.s. fails by BC II (independence + $\sum 1/n = \infty$).
- **Q2(b):** a.s. holds by BC I ($\sum 1/n^2 < \infty$, no independence needed).
- **Q2(c):** Harmonic series $\Rightarrow \mathbb{E}[S_n] \sim c \log n$; variance $= O(\log n)$; L^2 argument closes the probability convergence.